



UNIVERSITÉ
DE LORRAINE



IUT

Saint-Dié-des-Vosges

INSTALLATION OF A DEVELOPMENT STATION

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INTRODUCTION

As part of our SAE 1.03 Led Study, we conducted a project to explore and compare the current major operating systems, make a Linux installation guide and other small specifics.



1. PURPOSE AND MISSION :

The technical objective of our project is divided into two important parts:

Part 1: Comparative study of operating systems

The main purpose of this part was to determine the best operating system (OS) for a type of developer. We decided to choose the full-stack developer profession.

To achieve this goal, we conducted a comprehensive comparative study of the main operating systems: Linux, Windows and macOS.

This study is aimed primarily at full stack developers, whether they are beginners or experienced, as well as companies looking to choose the best development environment for their teams.

Our mission was to analyze each operating system according to the specific needs of a full stack developer, taking into account criteria such as:

- Compatibility with common development tools
- The ease of setting up the development environment
- Performance for development tasks
- The stability and reliability of the system
- Security
- Flexibility and customization

At the end of our in-depth comparative study of Linux, Windows and macOS operating systems, we had to determine the best choice for a full-stack developer. After careful analysis, we opted for Windows and here is why:

- *Broad software compatibility: Windows offers exceptional compatibility with a wide range of development tools, frameworks and technologies essential for full-stack development.*
- *Extended support: The large community of Windows developers and official Microsoft support make it easy to quickly resolve technical issues.*

- *Powerful integrated development environment (IDE): Visual Studio, the flagship Microsoft IDE, is particularly suited to full-stack development, offering advanced features for both front and back end.*
- *Ease of use: The intuitive Windows interface allows developers to focus on their work without having to manage complex system configurations.*

This diagram highlights the range of skills required for a full-stack developer and underlines the importance of a versatile operating system like Windows to effectively manage these different layers.

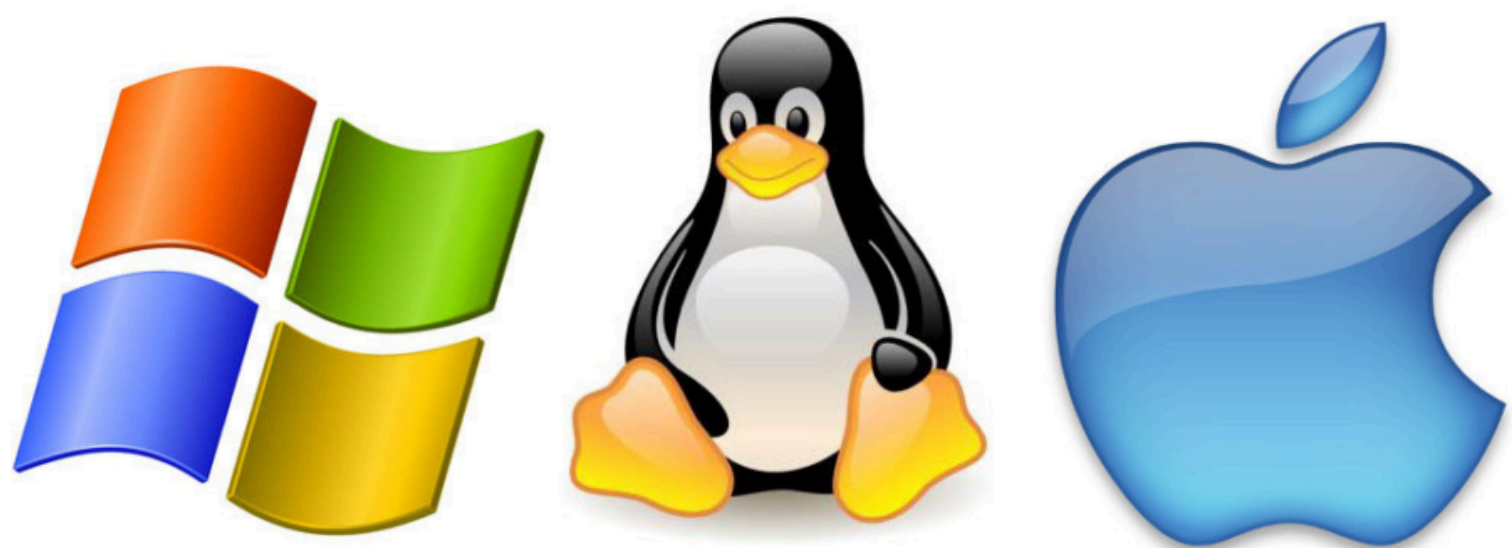
Part 2: Complete Linux Installation

The second important part of our project was to make a complete installation of Linux, explaining each step of the process and making the system fully functional. This part had the following objectives:

- *Provide a detailed guide for installing Linux*
- *Explain the different steps of configuration*
- *Make the operating system fully configured*

This practical approach complements the theoretical study by providing real-world experience in installing and configuring a Linux system.

To illustrate the working environment of a full-stack developer, we created a diagram of the typical software architecture of a complete web application



2. WORKING METHOD :

Our working method for this project was structured as follows:

Distribution of tasks :

Tas Tom :

- Creation of bootable key
- Explanation of the installation strategy
- Schematic of the software architecture encompassing the layers required for a complete web application
- Installation of Linux.

Cesar Lucas :

- Deep research and analysis on Linux, Windows and Mac OS
- Linux installation
- Choice of operating system
- Brief explanation of the full-stack developer job.

Work organisation :

- Creating a shared document to centralize our research
- Regular meetings to review progress and adjust our approach

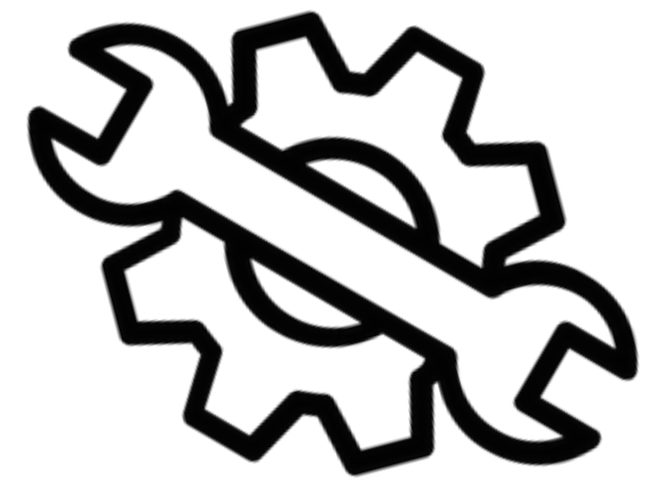
List of main tasks:

- Document search on each operating system (Linux, Windows, macOS)
- Analysis of the advantages and disadvantages of each system
- Writing individual sections for each OS
- Operating system comparison for full-stack development
- Creation of a schematic of the software architecture for a complete web application
- Creating Linux installation guide
- Compilation and formatting of the final document

We worked collaboratively, ensuring that each aspect of the project was covered in a balanced way. Regular communication has allowed us to maintain consistency in our approach and quickly resolve the difficulties encountered.



3. TECHNICAL TOOLS:



For this project, we utilized various hardware, software tools, and developed several key competencies: Hardware:

We used **two USB drives** and **a computer** for this project. These essential tools allowed us to create bootable media and perform the installation process. Software and Tools:

We employed **Rufus** to format a USB drive and make it bootable, which was crucial for the Linux installation.

We installed and configured **a Linux operating system**, gaining hands-on experience with this versatile platform.

Additionally, we utilized **the Office suite**, particularly Word's sharing feature, to enhance our collaboration and coordination.

Visio was used to create representative **diagrams**, improving our visual communication skills.

Developed Competencies: In **system installation and configuration**, we learned how to download and install a Linux operating system.

We improved our proficiency by **using Linux commands** for package and driver installation. We also gained experience in partitioning memory space, a critical skill for system management.

Our **device management skills** were enhanced through the preparation of bootable USB drives.

This process taught us about **creating portable, bootable media** for system installation and recovery.

We significantly improved our command-line skills, particularly in Linux. This experience has given us a deeper understanding of system operations and troubleshooting techniques.



4. DIFFICULTIES :

We have encountered some problems like:

1. Choosing the right operating system

Selecting an **appropriate operating system** for a **full-stack developer** involves considering various factors such as:

- Compatibility with development tools and software
- Performance and resource management
- Security features
- User interface and ease of use
- Community support and documentation availability

The decision often comes down to choosing between Windows, macOS, or a Linux distribution, each with its own strengths and weaknesses

2. Selecting the ideal work environment

This involves **setting up** the **development environment**, including:

- Integrated Development Environments (IDEs)
- Text editors
- Version control systems
- Package managers
- Terminal emulators
- Virtualization tools

The challenge lies in creating a cohesive and efficient workspace that caters to both front-end and back-end development needs

3. Understanding the needs of a full-stack developer

A full-stack developer **requires** a **diverse set of tools** and technologies, including:

- Front-end technologies (HTML, CSS, JavaScript, frameworks)
- Back-end languages and frameworks
- Database management systems
- API development tools
- Version control systems
- Cloud platforms and deployment tools

The difficulty lies in identifying and mastering the essential tools while staying updated with rapidly evolving technologies

4. Downloading and installing an OS on new storage

This **process involves**:

- Obtaining the correct OS image
- Verifying the integrity of the downloaded file
- Preparing the storage device (partitioning, formatting)
- Creating bootable media
- Configuring BIOS/UEFI settings
- Following the installation process

The challenge is in ensuring a smooth installation without corrupting existing data or encountering compatibility issues



5. Creating a bootable drive and storage space for the OS

This **step includes**:

- Selecting an appropriate USB drive or other bootable media
- Using tools like Rufus to create a bootable drive
- Partitioning the target storage device
- Ensuring sufficient space for the OS and user data
- Configuring the boot order in BIOS/UEFI

The difficulty lies in properly preparing both the bootable media and the target storage to ensure a successful OS installation

6. Creating an installation guide

Developing a comprehensive **installation guide** involves:

- Documenting each step of the process clearly
- Anticipating potential issues and providing solutions
- Including screenshots or diagrams for visual aid
- Ensuring the guide is accessible to users with varying technical expertise
- Keeping the guide up to date with any changes in the installation process

Creating a guide need to be detailed enough for novices and concise enough for expert

5. TO CONCLUDE :

The work was carried out in a positive atmosphere, and we consulted each other whenever necessary.

We gathered at each other's homes to work in the same environment, fostering collaboration and teamwork.

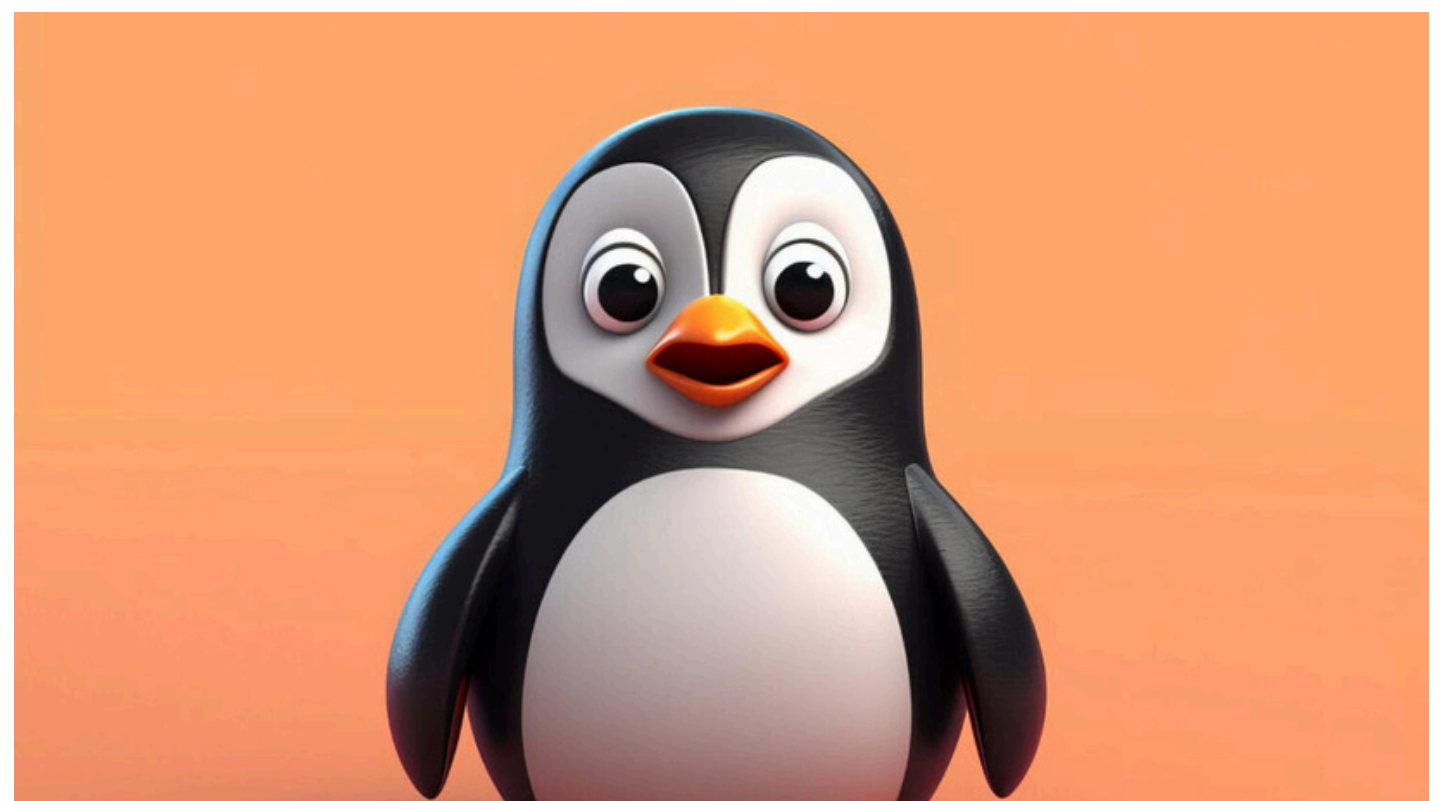
This was a simple yet enriching experience that allowed us to question certain approaches, particularly regarding the Linux installation process.

Overall, the SAE went very well. Most of the time, we were on the same page, and our viewpoints aligned in the same direction.

For this project, it might be preferable to work alone since the workload is not very heavy.

However, working as a team allows for error correction and provides opportunities to share and discuss project progress.

As the saying goes, "Alone we go faster, but together we go further." (Cory Booker)



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